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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

MUN SOCIETY
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# STUDY GUIDE
# UNITED NATIONS ECONOMIC AND SOCIAL COUNCIL

**Agenda 1:** Evaluating Frameworks for Adaptation Financing in Climate-Vulenrable States

**Agenda 2:** Addressing the Impact of Commercially-Induced Groundwater Depletion



Mumbai MUN'26

# Letter from R&D

Dear Delegates,
The Department of Research & Development, MUNSociety MPSTME, ecstatically welcomes all of you to the 13th edition of Mumbai MUN! Over the years, R&D has been guided by a singular ethos and vision: to make the Mumbai MUN conference a centre of meaningful discourse and debate on matters that rapidly shape the world we inhabit.

We are primarily tasked with formulating agendas and curating study guides that provide delegates with a comprehensive foundation before the conference. Our guides are the product of months of painstaking, human-driven research and production and are written with a *first-principles*, top-down approach.

Debate over matters of socio-political importance has taken centre stage in student life of late, with varying opinions and belief systems finding themselves continuously questioned. At Mumbai MUN 2026, as reflected by our theme "An Axiom Apart," we encourage delegates to bridge seemingly opposite truths and work together to creatively craft lasting solutions and partnerships.

With the *phare* that is our theme, our curations seek to foster meaningful and thought-provoking conversations on the topics that will define the global dynamics of the ever-changing world we live in. We seek to platform the perspectives that whittle away at the bureaucracy that barricades change.

The goal of any worthwhile MUN conference is to train the citizens of tomorrow to design, ideate, and create equitable world orders. The Research & Development team earnestly hopes Mumbai MUN 2026 will be a stepping stone for delegates to achieve that very goal.

Best Wishes,
Department of Research & Development,
MUN Society MPSTM

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# Agenda 1: Evaluating Frameworks for Adaptation Financing in Climate-Vulnerable States

## COMMITTEE OVERVIEW

The Economic and Social Council (ECOSOC) is one of the six principal organs of the United Nations, responsible for coordinating economic, social, and related work for UN specialised agencies and institutions. ECOSOC consists of 54 members, 18 of which are elected each year by the General Assembly for a three-year term.[1] Current members include 14 African States, 11 Asia-Pacific States, 6 Eastern European States, 10 Latin-American and Caribbean States, in addition to 13 Western European States. ECOSOC formulates policy advice addressed to member states and the UN system.

The council holds multiple meetings and sessions to discuss major economic, social and environmental policy issues. It can make recommendations for the purpose of promoting respect for, and observance of, human rights and fundamental freedoms for all as well as prepare draft conventions for submission to the General Assembly, with respect to matters falling within its competence.[2]

## MANDATE

The mandate of the Economic and Social Council (ECOSOC) is to coordinate the economic, social, cultural, and environmental work of the United Nations and promote global development.[3] ECOSOC acts as the main global forum for discussing international economic and social issues.

Its main objectives include higher standards of living, full employment, and conditions of economic and social progress and development; Solutions of international economic, social, health, and related problems; and international cultural and educational cooperation; Universal respect for, and observance of, human rights and fundamental freedoms for all without distinction as to race, sex, language or religion.

Additionally, it coordinates UN specialised agencies, oversees the implementation of agendas like the 2030 Sustainable Development Goals, supports principal UN bodies, and serves as a vital platform for NGO engagement.[4]

## AGENDA OVERVIEW

Vulnerable states suffer the worst impacts of global warming despite contributing the least to greenhouse gas emissions. The global adaptation finance gap keeps increasing, which leaves climate-vulnerable countries susceptible to debt. This gap leads to economic challenges like resource diversion, higher borrowing costs, as well as social impacts like forced mass displacement and food insecurity. Current international flows are able to meet only a fraction of actual needs for climate-vulnerable states. Although several frameworks and policies exist, a large majority of funds are not able to reach local communities. This agenda explores the causes of climate vulnerability and the impact of lack of funding, as well as the challenges faced by climate-vulnerable states.

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[1] United Nations. (n.d.-c). *Economic and Social Council Documentation*. United Nations. https://research.un.org/en/docs/ecosoc

[2] United Nations. (n.d.-b). *Chapter X: The Economic and Social Council (articles 61-72)*. United Nations. https://www.un.org/en/about-us/un-charter/chapter-10

[3] *Mandate | Economic and Social Council - the United Nations*. United Nations. (n.d.-a). https://ecosoc.un.org/en/annual-ministerial-review/mandate

[4] *ECOSOC brochure*. ECOSOC. (n.d.). https://ecosoc.un.org/sites/default/files/2023-06/ECOSOC_Brochure.pdf

## KEY CONCEPTS

### Climate Vulnerability

The Intergovernmental Panel on Climate Change (IPCC) defines climate vulnerability as the propensity or predisposition of a species to be adversely affected and its inability to cope with the negative effects of climate change.

Climate vulnerability has three central components:
(a) Exposure: It is the amount of climate change a species or a system faces (e.g. temperature changes, precipitation changes).
(b) Sensitivity: How severely a species or a system is affected when climate change takes place.
(c) Adaptive Capacity: The ability of a species to cope, recover and adapt to climate change through migration, evolution, etc.[5]

### Small Island Developing States (SIDS)

Small Island Developing States are a diverse group of states that are highly susceptible to climate change and face comparable sustainable development challenges due to their geographical disadvantages and socio-economic vulnerabilities. They are small in size and are located in remote regions of the world. Some of these states are dependent on the ocean for their resources, but in the case of external disasters, they become heavily reliant on imports and loans.[6] SIDS are divided into three groups: The Caribbean SIDS; The Pacific SIDS; and the Atlantic, Indian Ocean, and South China Sea (AIS) SIDS.[7] The United Nations recognises 57 SIDS, out of which 39 are sovereign states, and 18 are associate members.[8]

### Least Developed Countries (LDC)

The United Nations defines LDCs as countries that have low levels of income and face severe structural impediments to sustainable development. A country is categorised as an LDC if its Gross National Income is below $1,152 per capita, its score on the Human Asset Index is 60 or below, and its Economic Vulnerability Index is 36 or above.[9] The United Nations classifies 44 countries as LDCs, 32 of which are in Africa, 8 in Asia, 3 in the Pacific and 1 in the Caribbean.[10]

### Adaptation Finance

Adaptation Finance refers to helping people and countries adapt to the impact of climate change. It does so by building resilience to climate change impacts, from flood infrastructure to drought-resilient agriculture, early warning systems, and risk-transfer mechanisms like insurance. Adaptation seeks to reduce the risks posed by climate change while seeking to best position infrastructure and systems for the future. [11]

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[5] *Climate change vulnerability assessments*. Climate Change Vulnerability Assessments | Climate Action Tool. (n.d.). https://climateactiontool.org/content/climate-change-vulnerability-assessments

[6] United Nations. (n.d.). *About small island developing states | office of the high representative for the least developed countries, landlocked developing countries and small island developing States*. United Nations. https://www.un.org/ohrlls/content/about-small-island-developing-states

[7] Environment, U. (n.d.). *UNEP & Small Island Developing States (SIDS)*. UNEP. https://www.unep.org/topics/ocean-seas-and-coasts/small-island-developing-states/unep-small-island-developing-states

[8] United Nations. (n.d.). *About small island developing states | office of the high representative for the least developed countries, landlocked developing countries and small island developing States*. United Nations. https://www.un.org/ohrlls/content/about-small-island-developing-states

[9] *Inclusion in the LDC category | economic analysis and policy division*. Least Developed Countries Category. (n.d.). https://policy.desa.un.org/themes/least-developed-countries-category/inclusion-in-the-ldc-category

[10] *UN list of Least Developed Countries*. UN Trade and Development (UNCTAD). (n.d.). https://unctad.org/topic/least-developed-countries/list

[11] *Adaptation finance: The Challenge for investors and banks*. World Economic Forum. (n.d.). https://www.weforum.org/stories/climate-action/climate-adaptation-finance-the-challenge-for-institutional-investors-and-commercial-banks/

## KEY FRAMEWORKS

### Adaptation Fund

The Adaptation Fund was established in 2001 under the UN Framework Convention on Climate Change (UNFCCC) and its Kyoto Protocol; later also serving the Paris Agreement. The Adaptation Fund helps finance projects and programmes that help vulnerable communities in developing countries adapt to climate change. Actions are based on country needs, views and priorities. Upon its establishment, it was decided that the Adaptation Fund would be financed from the share of proceeds (SOP) on the clean development mechanism, project activities and other sources of funding, including voluntary contributions. Its primary funding comes from a 2% share of proceeds of the Certified Emission Reductions issued by the Kyoto Protocol's Clean Development Mechanism.[12]

### Special Climate Change Fund (SCCF)

The Special Climate Change Fund(SCCF), one of the world's first international climate adaptation finance instruments, was created at the 2001 Conference of the Parties (COP) to the UNFCCC to help vulnerable nations address the negative impacts of climate change.

The SCCF was created to finance projects relating to adaptation; technology transfer and capacity building; energy, transport, industry, agriculture, forestry and waste management; and economic diversification. The SCCF is managed by the Global Environment Facility (GEF) and operates simultaneously with the Least Developed Countries Fund (LDCF). Both funds have a mandate to serve the Paris Agreement.[13]

### Least Developed Countries Fund (LDCF)

The LDCF is an international fund that was created to help countries that have low levels of income adapt to the harmful effects of climate change. The LDCF was established in November 2001 under the UNFCCC.

It has one of the largest records of adaptation funding and has delivered more than $2.35 billion in total grants for climate adaptation projects since its establishment. The LDCF helps prepare and execute National Adaptation Programs of Action (NAPAs).[14]

### Green Climate Fund (GCF)

The Green Climate Fund is the world's largest international fund. It aims to support low-emission, climate-resilient development pathways in developing countries. The GCF is the largest provider of adaptation finance among climate funds. At least half of its adaptation funding is directed to the countries most vulnerable to the effects of climate change, including LDCs and SIDS. [15]

Established in 2010 by the 194 parties to the UNFCCC, the GCF's duty is to support drastic changes in climate mitigation and climate adaptation efforts. The GCF is a legally independent entity with a full-time Secretariat in Songdo, Republic of Korea, and is designed as an important multilateral fund that would promote a different model for fighting climate change.[16]

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[12] Adaptation Fund. (n.d.). https://unfccc.int/Adaptation-Fund

[13] *Special climate change fund - SCCF*. Global Environment Facility. (2025a, December 17). https://www.thegef.org/what-we-do/topics/special-climate-change-fund-sccf

[14] *Least Developed Countries Fund - LDCF*. Global Environment Facility. (2025a, December 17). https://www.thegef.org/what-we-do/topics/least-developed-countries-fund-ldcf

[15] *The World's Climate Fund for Developing Countries*. Green Climate Fund. (n.d.). https://www.greenclimate.fund/about

[16] *Green Climate Fund*. World Bank. (n.d.). https://fiftrustee.worldbank.org/en/about/unit/pim/fiftrustee/fund-detail/gcftf

### CVF V20's Joint Multi-Donor Fund (JMDF)

The CVF V20 Joint Multi-Donor Fund (JMDF) is a united financial system created to facilitate coordination among the respective member states of the Climate Vulnerable Forum (CVF) and Vulnerable Twenty (V20) Group of the Ministers of Finance of the CVF. Its goal is to carry out the overall aim of enhancing CVF and V20 members' capacities to deliver on the key issue of climate change, both in terms of climate change adaptation and mitigation. The JMDF was initiated in 2020 to support climate action and resilience in 74 climate-vulnerable countries.

Overall, the JMDF aims to contribute to the implementation of the UNFCCC, the Paris Agreement, as well as the UN Sustainable Development Goals (SDGs). In 2024, the CVF-V20 became an independent institution with the capacity to manage its own fund. This means that the JMDF now fulfils the function of receiving resources primarily from new bilateral donors (e.g., the Federal Republic of Germany, the People's Republic of Bangladesh, etc.) to increase climate prosperity.[17]

### National Adaptation Plan (NAP)

The NAP process was established in 2010 under the Cancun Adaptation Framework at the 16th Conference of the Parties (COP 16) to the UNFCCC. NAPs are formal government-led processes that let countries adapt and plan for the long-term effects of climate change. The objectives of the NAP process are:

(a) To reduce vulnerability to the impacts of climate change, by building adaptive capacity and resilience.

(b) To facilitate the integration of climate change adaptation, in a coherent manner, into relevant new and existing policies, programmes and activities, in particular development planning processes and strategies, within all relevant sectors and at different levels, as appropriate. Funding related to the formulation and implementation of NAPs is mandated to be provided through the Green Climate Fund, the Least Developed Countries Fund, and the Special Climate Change Fund.[18]

### Paris Agreement

The Paris Agreement is a legally binding international treaty on climate change. It was adopted by 195 Parties at the UN Climate Change Conference (COP21) in Paris, France, on 12 December 2015. It entered into force on 4 November 2016. Its core goals are:

(a) To keep the global average temperature increase to well below 2°C above pre-industrial levels, while pursuing efforts to limit the increase to 1.5°C.

(b) To provide financing to developing countries to mitigate climate change, strengthen resilience and enhance abilities to adapt to climate impacts.

(c)To periodically assess the collective progress towards achieving the purpose of this agreement and its long-term goals.[19]

The Paris Agreement works on a five-year cycle of increasingly ambitious climate action carried out by countries. Every five years, each country is expected to submit an updated national climate action plan known as a Nationally Determined Contribution (NDC). It also reiterates that developed countries should provide financial, technological, and capacity-building support to

[17] *CVF-V20 Joint Multi-Donor Fund*. CVFV20.Org. (2026, February 6). https://cvfv20.org/jointmultidonorfund/
[18] *National Adaptation Plans*. Unfccc.int. (n.d.). https://unfccc.int/national-adaptation-plans#aboutNAPs
[19] United Nations. (n.d.-b). *The Paris Agreement*. United Nations. https://www.un.org/en/climatechange/paris-agreement

help developing countries reduce emissions and adapt.[20]

## MAJOR ACTORS

### United Nations Framework Convention on Climate Change (UNFCCC)

The UNFCCC was adopted at the United Nations Headquarters, New York, on 9 May 1992.
The primary objective of this Convention is to achieve stabilisation of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system.

The level must be achieved within a time frame sufficient to allow ecosystems to adapt naturally to climate change, to ensure that food production is not threatened and to enable economic development to proceed in a sustainable manner. [21]

The commitments that all parties make, taking into account their common but differentiated responsibilities and their specific national and regional development priorities, objectives and circumstances, shall:

(a) Develop, periodically update, publish and make available to the Conference of the Parties national inventories of anthropogenic emissions by sources and removals by sinks of all greenhouse gases not controlled by the Montreal Protocol, using comparable methodologies to be agreed upon by the Conference of the Parties.

(b) Formulate, implement, publish and regularly update national and, where appropriate, regional programmes containing measures to mitigate climate change by addressing anthropogenic emissions by sources and removals by sinks of all greenhouse gases not controlled by the Montreal Protocol, and measures to facilitate adequate adaptation to climate change [22]

The developed country Parties and other Parties commit themselves specifically as provided for in the following:

(a) To promote progress, each of these Parties shall communicate, within six months of the entry into force of the Convention for it and periodically thereafter, to return individually or jointly to their 1990 levels these anthropogenic emissions of carbon dioxide and other greenhouse gases not controlled by the Montreal Protocol. This information will be reviewed by the Conference of the Parties, at its first session and periodically thereafter, in accordance with Article 7.[23]

(b) Calculations of emissions by sources and removals by sinks of greenhouse gases should take into account the best available scientific knowledge, including of the effective capacity of sinks and the respective contributions of such gases to climate change. The Conference of the Parties shall consider and agree on methodologies for these calculations at its first session and review them regularly thereafter

### Global Environment Facility (GEF)

The Global Environment Facility (GEF) is the world's largest multilateral fund for the environment. It supports countries' efforts to tackle the root causes of environmental degradation, advancing global goals for nature, climate, and pollution. The facility consists of 186

[20] *Paris Agreement*. Unfccc.int. (n.d.-b). https://unfccc.int/process-and-meetings/the-paris-agreement
[21] Unfccc.int. (n.d.). https://unfccc.int/process-and-meetings/united-nations-framework-convention-on-climate-change
[22] United Nations Framework Convention on Climate Change United Nations 1992. (n.d.-k). https://unfccc.int/resource/docs/convkp/conveng.pdf
[23] United Nations Framework Convention on Climate Change United Nations 1992. (n.d.-k). https://unfccc.int/resource/docs/convkp/conveng.pdf

member countries; there are developed and developing participants, both of which are donors to the trust fund.[24]

The fund is made available to developing countries and countries with economies in transition to meet the objectives of international environmental conventions and agreements.

Funds are implemented as targeted initiatives across five focal areas: biodiversity loss, chemicals and waste, climate change, international waters, and land degradation.

The GEF has been the largest public funder for the environment and has provided more than $27 billion in financing, primarily as grants, and mobilised another $155 billion for country-driven priority projects.

Other initiatives include collaborating and working with local communities, Indigenous Peoples, as well as voicing the youth.[25]

### Climate Vulnerable Forum and V20 Finance Ministers (CVF-V20)

Established in 2009, the Climate Vulnerable Forum and V20 Finance Ministers (CVF-V20) is an international partnership of countries highly vulnerable to a warming planet.

It serves as a South-South intergovernmental cooperation platform where progressing countries pursue their development objectives through exchanges of knowledge, skills, and resources for their individual and/or mutual benefit within and across regions.

It includes 74 member countries from Africa, Asia, the Caribbean, Latin America, the Middle East, and the Pacific, representing 1.7 billion people. Collectively, these nations contribute only 7.3% of global emissions, yet they face some of the most severe climate impacts. [26]

The CVF-V20 advocates development-positive climate actions which are built around the principles of prioritising long-term development, empowering vulnerable countries to achieve their development goals, and developing new approaches to climate financing.[27]

### Resilience and Sustainability Trust (RST)

The IMF's Resilience and Sustainability Trust (RST) helps low-income and vulnerable middle-income countries build resilience to external shocks and ensure sustainable growth, contributing to their longer-term balance of payments stability.

The IMF's mandate is to promote members' balance of payments stability, as well as foster global financial stability and economic growth.

The Poverty Reduction and Growth Trust (PRGT), established before the RST, aims to fund itself using earnings from its investments to provide concessional loans to its borrowers, including interest-free financing to the poorest countries.[28]

The RST offers loans, with a 20-year maturity and a 10½ year grace period during which no principal is repaid, to countries that are eligible for the PRGT, low-income countries, small states (population under 1.5 million) with per capita GNI below 25 times the 2021 IDA operational cutoff, and all middle-income countries with per capita GNI below 10 times the 2021 IDA operational cutoff.

[24] *Who we are*. Global Environment Facility. (n.d.). https://www.thegef.org/who-we-are
[25] *Who we are*. Global Environment Facility. (n.d.). https://www.thegef.org/who-we-are
[26] *History*. CVFV20.Org. (2026a, September 7). https://cvfv20.org/history/
[27] *History*. CVFV20.Org. (2026a, September 7). https://cvfv20.org/history/
[28] *IMF resilience and Sustainability Trust. (n.d.-e)*. *https://www.imf.org/en/topics/resilience-and-sustainability-trust*

### Financial Institutions

Financial institutions are companies involved in monetary transactions such as loans, deposits, investments and insurance.

Financial institutions serve as the first line of defence against risks of all kinds, with hundreds of millions of low-income and vulnerable clients across the world.

By adapting to reflect the new climate realities, these institutions can directly enable clients' personal adaptation and resilience strategies in ways that few others can. Financing investments that reduce risk and mitigate the impacts of shocks before they happen.

The inclusive finance sector can also allow global climate funds to enable small-ticket financing, which the current climate finance architecture cannot do because it was built to finance projects, not people.[29]

The inclusive financial services systems built over the last few decades can crowd in up to $1.5 trillion of private-sector capital for climate mitigation and adaptation that represents a significant opportunity to respond to and enhance economic activity.

However, Most Financial Service Providers (FSPs) still lack strategies, risk tools and products required to adapt their business models to climate change.

CGAP's research over the past three years reveals that institutions beginning to develop climate strategies are often focused on regulatory compliance and internal net-zero goals, rather than on embedding resilience and adaptation into their core strategy, operations, and product offerings.[30]

## CAUSES

### Geographical Conditions

Certain countries that are situated in regions with extreme geographical conditions like mountainous terrain with significantly high altitudes, hot deserts, polar regions and low-lying islands are much more vulnerable to climate change issues. They also have fewer resources available to prepare for climate emergencies and fund reconstruction efforts.[31]

Mountains receive more rainfall than low-lying areas because, as air is forced over the higher ground, it cools, causing moist air to condense and fall out as rainfall.

The higher the place is above sea level, the colder it will be. This happens because as altitude increases, air becomes thinner and is less able to absorb and retain heat.

The Horn of Africa, a dry area, and the Maldives, a low-lying island country, are already feeling the strain of climate change while producing relatively few emissions.[32]

Arid and semi-arid regions like Somalia, Ethiopia and Eritrea are facing multi-year droughts due to the 1.2° C increase in today's climate caused by greenhouse gas emissions.

The Maldives, the world's lowest-lying nation with 80% of total landmass lying just one meter above sea level, could become uninhabitable by

[29] *Built to adapt: Inclusive financial institutions in a changing climate*. CGAP. https://www.cgap.org/research/publication/built-to-adapt-inclusive-financial-institutions-in-changing-climate
[30] *Built to adapt: Inclusive financial institutions in a changing climate*. CGAP. https://www.cgap.org/research/publication/built-to-adapt-inclusive-financial-institutions-in-changing-climate
[31] *Environmental refugee*. Education. (n.d.-a). https://education.nationalgeographic.org/resource/environmental-refugee/
[32] *Environmental refugee*. Education. (n.d.). https://education.nationalgeographic.org/resource/environmental-refugee/

2050 due to sea level rise and environmental degradation.

The amount of sunlight that is absorbed or reflected by the planet depends on the Earth's surface and atmosphere. Dark objects and surfaces, like the ocean, forests, and soil, tend to absorb more sunlight. Light-colored objects and surfaces, like snow and clouds, tend to reflect sunlight. About 70 per cent of the sunlight that reaches the Earth is absorbed. Natural changes in the Earth's surface, like the melting of sea ice, have contributed to climate change in the past.[33]

### Lack of Provisions for Protective Infrastructure

Low-income countries are less able to deal with climate events because they often lack the institutional, financial or technological capacity to adapt effectively.

The impacts of climate change are already being felt across the world, and the frequency and severity of extreme events are set to increase, damaging infrastructure systems and disrupting the provision of services to end users and communities.

As a result, infrastructure needs to be prepared to withstand the impacts of existing and future climate change. Failure to address climate resilience will ultimately make infrastructure systems unsustainable and undermine their financial viability.

There is an urgent need to retrofit existing infrastructure or construct new infrastructure with the specific aim of reducing hazard exposure and increasing the adaptive capacity of the asset and the community it serves. An example of this is the construction of flood protection infrastructure to protect a community and its assets against future flood hazards. Nature-based Solutions (NBS), like mangrove forests or wetlands, can serve this purpose and help increase the adaptive capacity of existing assets and communities while simultaneously providing other co-benefits.[34]

### Poverty and Income Disparity

Higher temperatures are expected to impact globally on poverty and inequality, yet little cross-country analysis exists to quantify the effects.

A panel dataset of subnational poverty in 130 countries covering the past decade finds that a 1 °C increase in temperature causes headcount poverty increases of 0.63–1.18 percentage points.

These poverty estimates equal a projected increase in the global poor by 62.3–98.7 million people by 2030 compared with a scenario without climate change.

Poorer countries, particularly those in Sub-Saharan Africa, are more vulnerable, as are countries with higher agriculture shares in the economy. Estimates at the subnational level are larger than those using the country-level data, indicating that aggregated analysis may underestimate climate change risks. [35]

### Lack of Central Planning

Adaptation finance suffers from a lack of central planning, which leads to uncoordinated and delayed money flow. Out of 172 countries with national adaptation plans, 36 have plans that are at least ten years old.[36]

[33] Environmental Protection Agency. (n.d.). EPA. https://www.epa.gov/climatechange-science/causes-climate-change
[34] Climate resilient infrastructure handbook. (n.d.-b). https://gca.org/wp-content/uploads/2025/06/Climate-Resilient-Infrastructure-Handbook_20250613.pdf
[35] Dang, H.-A. H., Hallegatte, S., Nguyen, M. C., & Trinh, T.-A. (2025, December 22). *Impacts of global warming on subnational poverty and inequality*. Nature News. https://www.nature.com/articles/s41558-025-02516-6#Sec2
[36] Book.com/Unep. (2025, October 29). *Slow climate adaptation threatening lives and economies*. UN Environment. https://www.unep.org/news-and-stories/press-release/slow-climate-adaptation-threatening-lives-and-economies

Administrative delays and weak local market planning mean regions like South Asia receive only half of their allocated adaptation funds.[37]

Governments do not track private sector investments in adaptation. Private companies invest without a shared national goal. National goals rarely connect with local and district needs. Local governments often lack the data, staff, or clear rules required to turn national climate plans into real action.[38]

## IMPACTS

### Economic Impacts

#### Resource Diversion

Resource diversion refers to shifting natural and financial resources away from traditional uses to respond to climate change and shocks. Natural resources regulate the economic, social and cultural lives.

Global water security is increasingly threatened by the unequal distribution of water resources; to address this issue, worldwide water diversion programmes (WDP) have been implemented.

However, the reallocation of water to meet demand may reshape the regional water cycles.[39]

Food is the majority of waste that ends up in landfills; the methane emissions from them account for 10% of global greenhouse gas emissions and 14% of methane emissions.

The process of waste management is resource-heavy, as it requires waste management entities to collect, divert and process it.[40]

A $103 billion has been reported to be taken from aid funds intended for health, education and gender equality by the Group of Seven (G7), an informal international forum consisting of leading democracies and other rich countries.

These countries have made hollow use of definitions of additional funds to developing countries and have thus failed to provide climate finance.[41]

#### Higher Borrowing Costs

Higher borrowing costs refer to increased interest rates and bond yields for governments and firms when exposed to extreme climate change.

As found using a global subset of 86,000 bank loans, a higher climate vulnerability of a firm's host country leads to higher costs of borrowing. The effects of physical climate risk on loan pricing are particularly large if loans have long maturities and if borrowing firms are in financial distress.[42]

A one standard deviation increase in a firm's exposure to physical climate risk, measured as

[37] Canales, & Nella. (2023, November 10). *Three major gaps in climate-adaptation finance for developing countries*. SEI. https://www.sei.org/perspectives/three-major-gaps-in-climate-adaptation-finance-for-developing-countries/

[38] Swadana, W. A., & Garcia, M. (2026, June 29). *How to turn national adaptation plans into local action in Brazil, India and Indonesia*. World Resources Institute. https://www.wri.org/technical-perspectives/national-local-climate-adaptation-brazil-india-indonesia

[39] Huang, N., Li, J., Zhu, S., & Wang, D. (2026). Greenhouse Gas Emissions From Water-Diversion Projects: Implications for Water-Resource Allocation and Climate Feedbacks. *Engineering*. https://doi.org/10.1016/j.eng.2026.08.016

[40] Book.com/Unep. (n.d.). *Food waste breakthrough launches to help cities halve food waste, cut methane emissions and reduce hunger*. UN Environment. https://www.unep.org/news-and-stories/press-release/food-waste-breakthrough-launches-help-cities-halve-food-waste-cut

[41] Perry, J. (2022, June 23). *World's richest countries top-up climate finance with funds diverted from world's poorest*. CARE Climate Change. https://careclimatechange.org/worlds-richest-countries-top-up-climate-finance-with-funds-diverted-from-worlds-poorest/

[42] Physical climate risk and the pricing of bank loans - Sciencedirect. (n.d.-m). https://www.sciencedirect.com/science/article/pii/S0095069625001640

the climate vulnerability of a firm's host country, increases its cost of borrowing by 39 basis points.

For lower-income countries in Africa and Asia, the effect of nature degradation on borrowing costs can be up to three times that of the mean.

Many of these countries host global biodiversity hotspots and export nature-dependent products to the rest of the world.[43] As a result, natural degradation in these countries is likely to have a huge effect on economies globally.

### Increased Public Debt

As climate disasters increase in both frequency and severity, the impact of these climate shocks on sovereign debt defaults remains large.

Because debt typically rises after disasters via financing recovery and reconstruction costs, any combination of these channels can amplify sovereign risk and push climate-vulnerable countries toward default.[44]

Recent disaster episodes show how large the associated economic and fiscal losses can be. In the case of Hurricane Ivan, which struck Grenada in 2004, total losses were estimated at about US$1.1 billion, representing 200 per cent of GDP. As a result, debt levels spiralled from 80 per cent of GDP in 2003 to 95 per cent of GDP in 2004, and the country undertook a debt restructuring over 2004-2006.[45]

### Agricultural Losses

Crops are sensitive to climate change, disrupting weather patterns and damaging soil and water resources, and the reduction in yield threatens global food systems.

Staple crops are especially vulnerable to heat, as an increase in climate temperature leads to a significant decrease in yield. Each 1° C increase in the mean temperature reduces global staple crop yields by 6 to 3%

In many areas, farmers are experiencing longer dry seasons and heat waves, with the planet being 1.5° C hotter than pre-industrial levels. By 2100, it is estimated that global crop yields would rise by 24% if emissions continue to rise.[46]

### Reduced Productivity and Growth

More than 171 million students globally, or 1 in 10 from pre-primary to upper secondary age, had their schooling disrupted by climate hazards in 2025.[47]

Learning losses and higher school dropout rates can reduce workforce productivity, slow economic growth, and make it harder for countries to reduce poverty. The report finds that inaction could result in a potential loss of at least US$200 billion in lifetime earnings for students who experienced climate-related school disruptions in 2025.[48]

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[43] Physical climate risk and the pricing of bank loans - Sciencedirect. (n.d.-m). https://www.sciencedirect.com/science/article/pii/S0095069625001640

[44] IMF elibrary. (n.d.-f). https://elibrary.imf.org/view/journals/001/2026/158/article-A001-en.xml

[45] Climate shocks, debt defaults and investment ... - IMF Elibrary. (n.d.-f). https://elibrary.imf.org/view/journals/001/2026/128/article-A001-en.xml

[46] *Climate change cuts global crop yields, even when farmers adapt*. Stanford Doerr School of Sustainability. (2025, November 21). https://sustainability.stanford.edu/news/climate-change-cuts-global-crop-yields-even-when-farmers-adapt

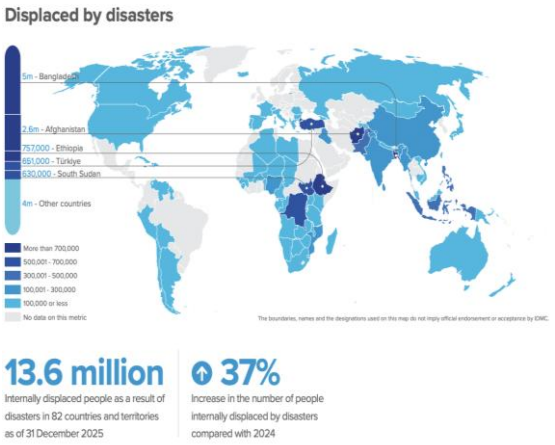
[47] More than 171 million students had their schooling disrupted by climate hazards in 2025 – UNICEF. (n.d.-e). https://www.unicef.org/armenia/en/press-releases/more-171-million-students-had-their-schooling-disrupted-climate-hazards-2025-unicef

[48] *More than 171 million students had their schooling disrupted by climate hazards in 2025 – UNICEF*. UNICEF Europe and Central Asia. (2026, September 10). https://www.unicef.org/eca/press-releases/more-171-million-students-had-their-schooling-disrupted-climate-hazards-2025-unicef

## Social Impact

### Forced Mass Displacement

Displacement and Climate Change are inherently intertwined. Vulnerable communities often don't have the resources to survive drastic climate variations. Due to a lack of financial support, these people have no choice but to flee. Out of the 82.2M Internally Displaced People (IDPs), 29.9 million were internally displaced as a result of disasters in 82 countries and territories as of 31 December 2025.[49] Developing countries are impacted the most by such displacement due to unpredictable weather conditions. Climate hazards like storms, floods, droughts, and heatwaves disrupt livelihoods and make it more difficult for displaced people to become self-sufficient.[50]




Internal Displacements in 2025 Due to Disasters
Source: Internal Displacement Monitoring Centre

### Food Insecurity

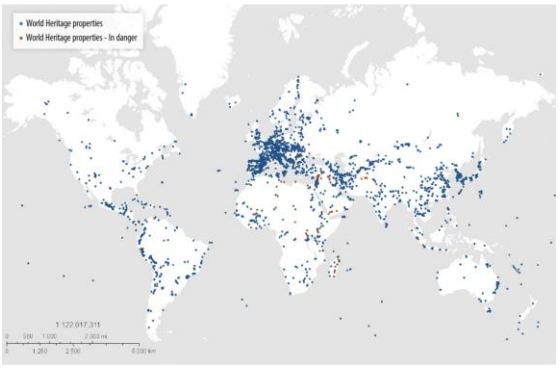
The negative impacts of climate change lead to irregular weather patterns, which in turn result in land degradation and a rise in temperature as well as sea levels.[51] This further escalates into higher costs for agricultural livelihoods and production. FAO's analysis of 78 post-disaster needs assessments in 48 developing countries spanning the 2003–2013 period shows that 25 per cent of all economic losses and damages inflicted by climate hazards in developing countries affect agricultural sectors. Climate change has both direct and indirect impacts on agricultural production systems. Direct impacts include effects caused by a modification of physical characteristics such as temperature levels and rainfall distribution on specific agricultural production systems. Indirect effects are those that affect production through changes in other species such as pollinators, pests, disease vectors and invasive species.[52]

### Culture and Heritage Loss

Climate change causes severe, often irreversible loss to global culture and heritage. Extreme weather conditions expose these important elements of our cultural identity to serious threats. Some direct impacts include structural decay, environmental pollution, and coastal erosion.[53] One of the biggest threats to marine heritage is climate change due to rising sea levels and erosion.[54] Indigenous and local communities face massive losses as they are forced to migrate

[49] *Global report on internal displacement*. IDMC. (2025). https://api.internal-displacement.org/sites/default/files/2026-05/IDMC-GRID-2026-Global-Report-on-Internal-Displacement.pdf
[50] United Nations. (n.d.). At the crossroads of crisis how the Climate Crisis Worsens displacement and conflict. United Nations. https://www.un.org/en/peace-and-security/at-the-crossroads-of-crisis
[51] United Nations. (n.d.-c). *The world's food supply is made insecure by climate change*. United Nations. https://www.un.org/en/academic-impact/worlds-food-supply-made-insecure-climate-change
[52] *Climate change and food security: Risks and responses*. FAO. (2015). https://openknowledge.fao.org/server/api/core/bitstreams/a4fd8ac5-4582-4a66-91b0-55abf642a400/content
[53] UNESCO. (n.d.-b). *Culture and Climate Action*. UNESCO. https://www.unesco.org/en/climate-change/culture
[54] *The impact of climate change on cultural heritage*. European Parliamentary Research Service. (2024, April). https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/762282/EPRS_BRI(2024)762282_EN.pdf

and have to sacrifice cultural traditions and practices.[55] Heritage sites face degradation and decay due to changing weather patterns.




Map of UNESCO World Heritage sites
Source: UNESCO

The impact of climate change on cultural heritage is also determined by the physical parameters, like the type of heritage involved: tangible(e.g. artefacts) or intangible(e.g. buildings), or immovable (e.g. buildings). [56]

### Unequal Impact on Marginalised Groups

Vulnerable groups such as low-income families, indigenous peoples, and minorities suffer unjust burdens due to climate change. These groups often rely on climate-based patterns for their livelihoods. Extreme weather ruins farms and fisheries, taking daily wages from rural workers.[57] Since their income depends on work done on a day-to-day basis, they often do not have the resources to build proper houses, which leaves them vulnerable to climate disasters. These marginalised communities usually do not have insurance and live in areas with underdeveloped public infrastructure. Thus, they have to rely on loans to cover their losses, as less than 17 per cent of climate finance reaches local levels.[58]

## CHALLENGES

### Massive Gaps for Adaptive Financing

Finance is a key enabling factor for adaptation, and financing constraints are a key determinant of the soft limits to adaptation, particularly in low-income to middle-income countries. Finance is therefore a key part of the UNFCCC negotiations.

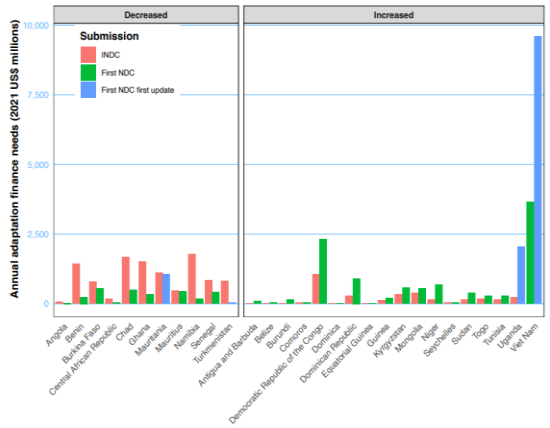
The estimated adaptation needs in USD. This is expressed as a range, based on two methods: a modelled estimate of the additional costs of adaptation for all developing countries, based on global sectoral models with national-level resolution, and an analysis of adaptation finance needs reported in developing countries' NAPs and NDCs, extrapolating these data to all developing countries.

The first step in estimating the adaptation finance gap is to estimate the costs of adaptation. In simple terms, this can be assessed by estimating the current and future economic impacts of climate change, then assessing the costs of reducing these impacts (the reduction being the benefit of adaptation).

However, there is a trade-off involved over how much adaptation to undertake, and on the level of residual damage costs after adaptation. This reflects the fact that adaptation is rarely completely effective, and that it usually becomes less cost-effective to deliver higher levels of risk reduction. This residual damage (after

[55] United Nations. (n.d.-b). *Climate change for indigenous peoples*. United Nations. https://www.un.org/development/desa/indigenouspeoples/climate-change.html
[56] *The impact of climate change on cultural heritage*. European Parliamentary Research Service. (2024, April). https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/762282/EPRS_BRI(2024)762282_EN.pdf
[57] Mishra, S. (2025, May 5). *Low-income communities bear the brunt of climate change*. Earth.Org. https://earth.org/climate-changes-unequal-burden-why-do-low-income-communities-bear-the-brunt/
[58] *Locally implemented or locally led? Tracking Finance for Community*. IIED. (2025, November). https://www.iied.org/sites/default/files/pdfs/2025-11/22674iied.pdf

adaptation) closely relates to the concept of loss and damage. [59]




Comparison of adaptation finance needs in initial and updated NDC submissions of developing countries. Source: UNFCCC

## Low Private Sector Involvement

New research on the potential for the private sector to deliver adaptation finance, commissioned by the Alliance and presented by Paul Watkiss, showed that private finance meets only 3% of needs and is mostly focused on a small number of sectors such as agriculture and water. [60]

Private investors may hesitate to invest in adaptation due to several significant barriers, including some that apply to climate projects in general, such as mispricing of natural resources and distortionary subsidies. The three challenges below are especially relevant to adaptation projects.

Adaptation projects may be perceived as riskier due to the uncertainty and complexity of climate impacts, and often result in public benefits rather than direct financial returns.

Investors may contend with limited access to information on climate impacts, future risks and likely adaptation outcomes. The impact of key approaches such as ecosystem-based adaptation has not been systematically measured; nor has the full range of potential environmental and social benefits been monetised and calculated. This makes it difficult to reliably calculate returns on investment and make informed investing decisions.

Most adaptation projects are inherently long-term, taking 10-20 years to implement. It is hard to make the business case for potentially large upfront costs today set against relatively long payback times. In addition, adaptation projects often have relatively small ticket sizes (around $30-$50 million), which may not appeal to traditional investors.[61]

## The Baseline Deficit Problem

The baseline deficit problem refers to a structural state where a region lacks the foundational infrastructure, institutional capacity, and socio-economic resilience required to meet standard, present-day development benchmarks required to take loans from major actors such as the CVF and V20. This sets back the particular region, as the funds never reach it in the first place.

Between 2010 and 2014, cities received less than 5% of global adaptation finance despite containing more than half of the world's population. [62]

Development finance institutions (DFIs), which provide non-commercial-based funding for economic and infrastructure development, are

[59] Adaptation finance gap update 2023. (n.d.-a). https://unfccc.int/sites/default/files/resource/Finance_Gap_Update.pdf

[60] *The role of the private sector in financing climate adaptation* . Zurich Climate Resilience Alliance. https://zcralliance.org/blogs/the-role-of-the-private-sector-in-financing-climate-adaptation/

[61] Choi, E., Jang, E., & Laxton, V. (2023, October 5). *What it takes to attract private investment to climate adaptation*. World Resources Institute. https://www.wri.org/insights/private-sector-climate-adaptation-finance

[62] An analysis of urban climate adaptation finance. (n.d.-b). https://www.climatepolicyinitiative.org/wp-content/uploads/2021/02/An-Analysis-of-Urban-Climate-Adaptation-Finance.pdf

constrained by their mandates and balance sheets from investing directly in cities, and international DFI financing processes are generally designed for national-level government recipients.

Cities in developing countries rely on revenues from property, income, sales, and fuel taxes, which lead to limited budget capacity outside of transfers from national governments or external finance institutions.

### Reliance on Loans over Grants

Climate finance is largely delivered through loans and equity, instruments that favour projects with clear financial returns. Because most adaptation projects do not generate steady revenues and rely on grants or concessional finance, this debt-heavy structure of climate finance systematically favours mitigation and leaves adaptation underfunded. Additionally, adaptation is also disadvantaged by the way finance gets delivered.[63]

Over 60% of adaptation finance is provided as loans rather than grants, increasing debt burdens for already vulnerable nations and limiting their ability to invest at large scale.[64]

## CASE STUDIES

### Viqueque Municipality

Located in Timor-Leste, this region faces significant development and climate-related challenges like irregular rainfall patterns, seasonal droughts, and rising sea levels.[65] The GCF and UN Development Programme carried out the project "Safeguarding Rural Communities and their Physical Assets from Climate-Induced Disasters".[66] This project is being implemented in stages across the region to strengthen the resilience of rural communities, protect important physical infrastructure, and reduce the risk of climate-induced disasters through adaptation measures.

Four major climate-resilient infrastructure interventions have already been carried out. In Suco Ahic, Administrative Post of Lacluta, a 3.5-kilometre rural road has been rehabilitated. In Suco Matahoi and Suco Uaitame, Administrative Post of Uatolari, the Belia 1 and 2 irrigation channels have been fully rehabilitated. The 1.95km system now delivers irrigation water to 2,117 citizens, essentially increasing agricultural production. Equipped with 11 water control gates, footbridges, gabion cascades, and erosion-control vegetation along its banks, the channel has reduced conflict over water access, stabilised production cycles, and enabled farmers to increase yields and income.

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[63] An analysis of urban climate adaptation finance. (n.d.-a). https://www.climatepolicyinitiative.org/wp-content/uploads/2021/02/An-Analysis-of-Urban-Climate-Adaptation-Finance.pdf

[64] United Nations. (n.d.). *Finance & Justice*. United Nations. https://www.un.org/en/climatechange/raising-ambition/climate-finance#:~:text=Doubling%20finance%20for%20countries%20to%20adapt%20to%20climate%20impacts&text=Building%20storm-resistant%20housing,%20planting,international%20public%20adaptation%20finance%20flows.

[65] *PreventionWeb*. PreventionWeb.net: the knowledge platform for disaster risk reduction. (2023, October 11). https://www.preventionweb.net/

[66] *2023 achievements of the Climate Resilience Project in Timor-Leste*. UNDP. (n.d.). https://www.undp.org/timor-leste/publications/2023-achievements-climate-resilience-project-timor-leste



A newly constructed water gate channels irrigation water from the main canal to nearby rice fields. Source: UNDP

The Green Climate Fund Board has approved US$22.9 million for funding, while the Government of Timor-Leste and the UNDP have provided US$36.687 million and US$400,000 respectively, for Timor-Leste to advance its adaptation priorities while focusing on more resilient rural infrastructure and strengthening local infrastructure development planning processes.[67]

### California

California wildfires cause massive financial damage due to infrastructure destruction as well as business interruption. The 2025 Los Angeles wildfire disaster alone led to $95 billion and $164 billion in damages, with insured losses estimated at $75 billion.[68] Low-income and immigrant workers like gardeners, cleaners, and service staff, who often lack institutional protection, were among the most affected, losing jobs or income as homes and workplaces were destroyed.[69] The destruction of housing intensified an already severe affordability crisis, pushing rent upwards. According to the California Department of Housing and Community Development (HCD) Regional Housing Needs Allocation (RHNA) for its 6th cycle, from October 2021 to 2029, L.A. County is required to construct 812,000 units to meet housing affordability goals. Midway through this cycle, only 71,000 units have been permitted, completing only 8.8% of the target. The wildfires have further exacerbated the already critical issue of insufficient housing supply in L.A.[70]

Federal partners initially made over $2 billion available through housing assistance and SBA low-interest disaster loans. Survivors have experienced prolonged delays as long-term recovery appropriations from the federal government remain unfulfilled after more than 18 months.[71]

Due to shortfalls in federal funding and escalating disaster costs, most of the responsibility for adaptation financing lands on state and local resilience programs. Community-driven initiatives like FireAid distributed $75 million across two phases for immediate emergency aid, mental health services, and housing stability.

### Nepal

On 26th August 2026, Nepal experienced a catastrophic flood triggered by a glacier

[67] *Safeguarding rural communities and their physical assets from climate induced disasters in Timor-leste*. UNDP Climate Change Adaptation. (n.d.-b). https://www.adaptation-undp.org/projects/safeguarding-rural-communities-and-their-physical-assets-climate-induced-disasters-timor

[68] Li, Z. (2025, February 13). *Economic impact of the Los Angeles wildfires*. PreventionWeb. https://www.preventionweb.net/news/economic-impact-los-angeles-wildfires

[69] The invisible costs of wildfire disasters in 2025. UNDRR. (2026a, January 13). https://www.undrr.org/news/invisible-costs-wildfire-disasters-2025

[70] Li, Z. (2025, February 13). *Economic impact of the Los Angeles wildfires*. PreventionWeb. https://www.preventionweb.net/news/economic-impact-los-angeles-wildfires

[71] Guardian News and Media. (2026, August 28). *Los Angeles Fire Survivors Face Record Wait for federal disaster aid under Trump*. The Guardian. https://www.theguardian.com/us-news/2026/aug/28/los-angeles-wildfire-survivors-federal-aid

collapsing. Nepal had set up early warning systems and adaptation measures that help reduce impacts from more conventional and forecastable floods. Despite such measures being in place, the magnitude of the calamity was beyond the predictive limits of existing risk reduction methods.[72]



Devastation in northern Nepal from the August 26 disaster
Source: UNICEF

Historically, Nepal has always been susceptible to loss and damage due to climate change. More than 80% of Nepal's residents experience climate-related disasters while two-thirds of the land is classified as climate-vulnerable. Nepal frequently experiences floods, landslides, droughts, glacial lake outburst floods (GLOFs), and heatwaves, leading to immense loss and damage (L&D) to the economy and infrastructure.[73]

Now, Nepali authorities have called for global efforts to tackle climate change, saying the country is one of the lowest contributors to greenhouse gas emissions yet one of the most vulnerable places to the effects of global warming.[74] Nepal has approached the UN Loss and Damage Fund, established in 2022 to assist developing countries that are particularly vulnerable to the adverse effects of climate change.[75] It seeks to claim roughly $20 million from this climate fund, though the mechanism faces severe funding gaps globally.[76]

## CONCLUSION

Adaptation is the primary focus in countries that face severe vulnerability to climate fallout, as is the case for many SIDs and LDCs. The current global funding architecture is fundamentally misaligned with actual climate needs because the system favours loans and mitigation over life-saving resilience.[77] Developing nations face an adaptation gap of US$310–365 billion per year; thus, shifting towards decentralised and grant-based models for adaptation funding can help close the gap for climate-vulnerable countries. ECOSOC can provide a platform for countries to meet and address such issues. A global effort is needed to implement ambitious climate action in view of accelerating climate impacts.[78]

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[72] *Rapid warming in the Himalaya exacerbates geohazard cascades beyond adaptation limits*. World Weather Attribution header. (2026, September 17). https://www.worldweatherattribution.org/rapid-warming-in-the-himalaya-exacerbates-geohazard-cascades-beyond-adaptation-limits/

[73] *Status of economic loss and damage caused by climate change in nepal: Patterns, impacts, and policy imperatives*. The Loss and Damage Collaboration. (2025). https://www.lossanddamagecollaboration.org/resources/status-of-economic-loss-and-damage-caused-by-climate-change-in-nepal-patterns-impacts-and-policy-imperatives

[74] Ewe, K. (2026, September 17). *Nepal to use new flood study to claim millions from UN Climate Disaster Fund*. BBC News. https://www.bbc.com/news/articles/cmlyrzve5pwzo

[75] United Nations. (2026, September 9). *Flood-ravaged Nepal calls for climate justice | UN news*. United Nations. https://news.un.org/en/story/2026/09/1168305

[76] Ewe, K. (2026, September 17). *Nepal to use new flood study to claim millions from UN Climate Disaster Fund*. BBC News. https://www.bbc.com/news/articles/cmlyrzve5pwzo

[77] *Reforming Global Finance for Climate-resilient development | UN Trade and Development (UNCTAD)*. UNCTAD. (2025, November 4). https://unctad.org/news/reforming-global-finance-climate-resilient-development

[78] Programme, U. N. E. (2025, October 1). *Adaptation gap report 2025*. UN Environment Document Repository Home. https://wedocs.unep.org/items/b547996e-14ee-4f1c-a6d4-b811dd373ae9

# Agenda 2: Addressing the impact of commercially induced groundwater depletion

## AGENDA OVERVIEW

ECOSOC stands to address social and economic issues; the depletion of groundwater aquifers, especially by the private sector, is a growing global concern. Groundwater accounts for 99% of all fresh water on Earth, and around 70% of aquifers are said to be in long-term decline.

The impacts of commercially induced extraction sit alongside global emergencies such as a drinking water crisis and food production collapse. Thus, Legal frameworks and treaties are implemented among aquifer states sharing transboundary aquifers to prevent over-extraction and aid in monitoring the water level and the geographical condition of the aquifer.

The challenges faced while addressing impacts of groundwater depletion include factors such as the persistence of the private sector, uncharted extraction data and natural causes such as environmental precarity.

## KEY DEFINITIONS

### Aquifers

An aquifer is a saturated geological formation in which groundwater flows with ease.
Generally, there are two types of aquifers: confined, having an impenetrable layer of rock above them, and unconfined, lying below a permeable layer or soil.

### Transboundary Aquifers

Transboundary aquifers are aquifers that cross political borders. The UN has mapped about 426 transboundary aquifers worldwide.




Transboundary aquifers of the world
Source: UNESCO

### Aquifer State

Aquifer State means a State in whose territory any part of a transboundary aquifer or aquifer system is situated. Nearly all non-island countries are identified as aquifer states which share transboundary aquifers.

## KEY FRAMEWORKS

### The UN Draft Articles on the Law of Transboundary Aquifers (2008)

Adopted on December 11, 2008, by the UN General Assembly (UNGA), the UN Draft Articles on the Law of Transboundary Aquifers are a set of nineteen international legal articles dictating the governance of shared underground freshwater resources. The articles cover the utilisation of aquifers, management of activities that could possibly affect aquifer systems, and the protection and preservation of international aquifers. Through their adoption by the UNGA and the International Law Commission (ILC), the articles represent the first global, multilateral effort towards codifying transboundary aquifer management.

The Draft Articles contain multiple provisions focused on the manner of aquifer utilisation and extraction. Article 4 mandates Aquifer States to utilise aquifers in an “equitable and reasonable”

manner, directing these states to maximise the benefits of the water contained in the aquifer while accounting for the long-term needs of the states involved. Article 5 sets down specific guidelines for Aquifer States to plan aquifer utilisation around, highlighting the need to be cognizant of both the socio-economic well-being of Aquifer State populations, as well as the environmental sustainability of the aquifer being used.[79]

The Draft Articles also include measures to promote the protection and sustainable development of aquifers. Articles 10 to 15 mandate Aquifer States to preserve aquifer ecosystems, reduce pollution, identify recharge zones and monitor aquifers to ensure the environmentally friendly use of groundwater resources. Further articles of the document contain provisions for seamless data sharing between Aquifer States and joint planned activities for aquifer management, along with guidelines for conduct during emergencies and armed conflicts, promoting diplomatic collaboration and partnership.[80]

However, critics of the document argue that the Draft Articles' declaration that Aquifer States have sovereignty over the section of aquifers located within their territory is a dangerous deviation from traditional freshwater law, stating that this view encourages nations to act in self-interest rather than collaborate on groundwater-related ventures.[81]

### The Guarani Aquifer Agreement (2010)

The Guarani Aquifer System (GAS) is a Transboundary Aquifer (TBA) shared by four countries in Latin America: Argentina, Brazil, Paraguay, and Uruguay. It is considered one of the largest freshwater reservoirs in the world. The GAS provides a broad legal framework under which the 4 countries must manage the GAS. The Guarani Aquifer Agreement mentions specifically that it is 'Taking into account the Resolution 63/124 of the United Nations General Assembly on the Law of Transboundary Aquifers'.

The analysis of the Guarani Aquifer Agreement leads to the following observations:

(1) Sovereignty can be considered a key management principle, but only in the context of applicable international law.

(2) Within the context of the obligation not to cause significant harm and to sustainably use the GAS.

(3) There is a common acknowledgement of the need to advance scientific knowledge on the GAS and to identify critical areas, both of which require built-in flexibility in the future legal regime for the management of the GAS.

(4) The future institutionalisation of the Guarani Aquifer Agreement lies in the La Plata River Basin Treaty, which will also play a role in the dispute settlement mechanism of the Guarani Aquifer Agreement.[82]

Current resource exploitation exceeds 1.0 cubic kilometre per year ($km^3/a$): 93.6% in Brazil, 2.8% in Uruguay, 2.3% in Paraguay, and 1.3% in Argentina. Some 80% of the total is used for public water supply, 15% for industrial processes and 5% by geothermal spas. Although this water

[79] Draft articles on the Law of Transboundary Aquifers (2008). (n.d.-g). https://legal.un.org/ilc/texts/instruments/english/draft_articles/8_5_2008.pdf
[80] Draft articles on the Law of Transboundary Aquifers (2008). (n.d.-g). https://legal.un.org/ilc/texts/instruments/english/draft_articles/8_5_2008.pdf
[81] McCaffrey, S. C. (2011). The International Law Commission's flawed Draft Articles on the Law of Transboundary Aquifers: the way forward. *Water International*, *36*(5), 566–572. https://doi.org/10.1080/02508060.2011.597094
[82] The guarani aquifer system: From a beacon of hope to a question mark in the governance of transboundary aquifers - sciencedirect. (n.d.-b). https://www.sciencedirect.com/science/article/pii/S2214581817303464

extraction is essential, it should be considered that it is still a small fraction of its full capacity.[83]

### The Franco-Swiss Genevese Aquifer Convention (1978)

Lake Geneva and Genevese groundwater provide drinking water to nearly 700,000 inhabitants of Geneva and the neighbouring French region. The groundwater, which is shared by the commune of Geneva, Switzerland, and the department of Haute-Savoie (Upper Savoy), in France, is exploited on both sides of the border due to ten water catchment wells in Switzerland and four in France.

Between 1940 and 1960, groundwater management of the Genevese aquifer was very close to the average sustained yield (7.5 M $m^3$/year). The groundwater level slowly fell without serious effect. Between 1960 and 1980, the aquifer became overused, with withdrawal rates of up to 14 million $m^3$/year in 1971, almost twice its potential yield. This over-pumping lowered the groundwater level by more than 7 m in 20 years, depleting total groundwater storage by about one third.[84]

This was when the first technical discussions took place both in France and in the canton of Geneva to find solutions that would curb the overexploitation of Genevan groundwater. In an effort to preserve this shared resource, political leaders, at the time, began to consider artificial groundwater recharge. [85]

## MAJOR ACTORS

### Private Enterprises

#### Commercial Bottled Water and Beverage Manufacturers

Bottled water has proved to be an important source of water consumption. While bottled water is used as an end product, it also contributes to other commercially produced beverages. Studies over the past decades show that Coca-Cola and Nestlé, respectively, use up to 1.95 litres and 4.1 litres of water to produce 1 litre of their final product. Various multinational corporations, such as Danone and Nestlé, extract about 3 million to 12 million litres of water per day from several regions around the globe. Thus, bottled water production is heavily dependent on groundwater sources, or lakes and reservoirs. Moreover, due to the requirements of the beverage industry, the retrieval of water from these sources further depletes them.[86]

#### Data Centre Operators

As AI systems grow more powerful and their deployment expands, so do their environmental footprints. Advanced models require immense computational resources, driving high energy consumption and associated carbon emissions, water withdrawals, and land use impacts. Data centres, which host most AI systems, often rely on water for cooling, sometimes withdrawing millions of litres per day. In many cases, these withdrawals occur in regions already facing drought or groundwater depletion, amplifying both environmental and social stress.

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[83] Gonçalves, R. D., Teramoto, E. H., & Chang, H. K. (2020). Regional Groundwater Modeling of the Guarani Aquifer System. *Water*, *12*(9), 2323. https://doi.org/10.3390/w12092323

[84] The genevese transboundary aquifer (Switzerland-france): The secret of 40 years of successful management - sciencedirect. (n.d.-j). https://www.sciencedirect.com/science/article/pii/S2214581817302665

[85] Cobos, G. de L. (2018, December). *The genevese transboundary aquifer (Switzerland-france): The secret of 40 years of successful management - sciencedirect*. Science Direct. https://www.sciencedirect.com/science/article/pii/S2214581817302665

[86] Global Bottled Water Industry: A review of impacts and trends. (n.d.-d). https://collections.unu.edu/eserv/UNU:9106/BottledWater_Report_Final_-compressed.pdf

Even when some withdrawn water is returned, large-scale withdrawals can strain aquifers and river systems, particularly in arid or groundwater-depleted regions. [87] Data centres may consume up to 110 million gallons of water in a year, while larger data centres can consume up to 1.8 billion gallons in a year.

As the number of data centres increases, developers are increasingly tapping into surface and underground aquifers to cool their facilities.[88]

### Agricultural Sector

Currently, 70% of global groundwater withdrawals, and even more in arid and semi-arid regions, are used in the agricultural production of food, fibres, livestock and industrial crops. Moreover, an estimated 38% of the lands equipped for irrigation are serviced by this resource.[89]

Groundwater depletion due to intensive withdrawals for agriculture is becoming an issue of increasing concern in certain areas, where it threatens to undermine food security, basic water supply, climate resilience, and the environmental integrity of groundwater-dependent wetlands and watercourses.[90]

### Other Industries

Industries that withdraw groundwater include manufacturing, mining, oil and gas, energy generation, engineering, and construction. Many production processes need a large amount of water for washing and cleaning their products at the end of production.

Mining industries not only deplete groundwater supplies but also pollute them. Heavy metals exposed during rock excavation contaminate the water supply.

Energy production processes use groundwater for various purposes such as cooling thermal generators and steam production in coal-based generation. These processes frequently deplete water from aquifers, modify the regional flow system, and reduce resource availability for other users.

All these other industries account for 19% of global freshwater withdrawals, including groundwater.[91]

## IMPACTS

### Food Production Collapse

Water scarcity drives food prices higher by reducing the amount of food available in markets while simultaneously raising the cost of producing what remains. Agriculture uses the largest share of global groundwater, using up to 40% of groundwater for consumptive irrigation.[92] Major food-producing regions like the Ganges Plains in India, the High Plains in the United States, and the North China Plain are

[87] *Environmental cost of artificial intelligence: Carbon, water, and land footprints*. United Nations University. (n.d.). https://unu.edu/inweh/collection/environmental-cost-of-AIs-Enrgy-Use-Carbon-water-and-land-footprints

[88] Environmental and Energy Study Institute (EESI). (n.d.). *Data Centers and water consumption*. EESI. https://www.eesi.org/articles/view/data-centers-and-wcontaminateater-consumption

[89] *Agriculture*. UNESCO.org. (n.d.). https://www.unesco.org/reports/wwdr/2022/en/agricu,,modifyreduceprocessesandcoal-basedlture

[90] *Groundwater is invisible, but its impact is visible everywhere*. RTP-RNE-WS. (2022, April 21). https://www.fao.org/platforms/water-scarcity/Knowledge/knowledge-products/detail/groundwater-is-invisible-but-its-impact-is-visible-everywhere/en

[91] *Industry*. UNESCO.org. (n.d.-b). https://www.unesco.org/reports/wwdr/2022/en/groundwater-and-industry

[92] *Groundwater conservation and food security: Navigating pathways to sustainable agriculture*. Ifpri.org. (n.d.). https://www.ifpri.org/blog/groundwater-conservation-and-food-security-navigating-pathways-to-sustainable-agriculture/

severely overexploited.[93] Global production of staples like rice and wheat declined at 1.8% and 1.5%, respectively, driving up market prices.[94]

### Drinking Water Crisis

Global drinking water shortages are worsening as underground aquifers deplete faster than natural rainfall can refill them.[95] Groundwater supplies drinking water to half of the global population and serves as the primary source for most rural communities.[96]

As groundwater levels further decrease, wells run dry, sacrificing local water access. Groundwater pumping may lead to the release of geogenic pollutants (e.g. arsenic and fluoride) present naturally in groundwater, making it unsafe to consume.[97]

### Land Subsidence

Land subsidence occurs when there is a loss of support below ground. This is most often caused by human activities, mainly from the overuse of groundwater, when the soil collapses, compacts, and drops.[98] Groundwater fills minuscule spaces between sand, silt, and clay, which helps it hold the weight of the land above.

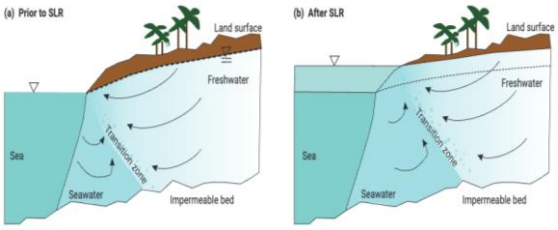
Excessive pumping depletes underground aquifers and causes subsidence. This also leads to reduced total storage capacity of the aquifer system. Land subsidence threatens inland urban areas by increasing flood risk and damaging infrastructure.[99]

### Seawater Intrusion

Excessive pumping in coastal areas can cause saltwater to move inland and upward, resulting in saltwater contamination of the water supply.[100] Coastal aquifers are under pressure from growing coastal populations and are increasingly vulnerable to seawater intrusion, which threatens both the quantity and quality of available water.

Seawater intrusion causes increased salinity of water, which is not suitable for human consumption. This also threatens agricultural production by decreasing soil fertility and crop yield.[101] Seawater intrusion can also be caused by land subsidence due to sinking of land.

Impact of sea level rise (SLR) on seawater intrusion in a sloping unconfined coastal aquifer system
Source: UNESCO




[93] *Alleviating groundwater depletion while realizing food security for sustainable development - sciencedirect*. Science Direct. (n.d.). https://www.sciencedirect.com/science/article/abs/pii/S0959652623005097
[94] Perez, N., Singh, V., & Ringler, C. (2024, June 14). *Ending groundwater overdraft without affecting food security*. Nature News. https://www.nature.com/articles/s41893-024-01376-w
[95] *World enters "era of Global Water Bankruptcy": UN scientists formally define new post-crisis reality for billions*. United Nations University. (n.d.). https://unu.edu/inweh/news/world-enters-era-of-global-water-bankruptcy
[96] *Groundwater, making the invisible visible*. UNESCO.org. (n.d.). https://www.unesco.org/reports/wwdr/2022/en
[97] UNESCO. (2022). *Groundwater: Making the invisible visible*.
[98] *Groundwater overuse and depletion*. Groundwater Foundation. (2022, September 11). https://groundwater.org/threats/overuse-depletion/
[99] *Land subsidence in coastal cities*. UNESCO. (n.d.). https://www.unesco.org/en/iggp/igcp-projects/663
[100] *Groundwater overuse and depletion*. Groundwater Foundation. (2022, September 11). https://groundwater.org/threats/overuse-depletion/
[101] Nolte, A. (2026, April 14). *Coastal groundwater-level trends reveal global susceptibility to seawater intrusion*. Nature News. https://www.nature.com/articles/s44221-026-00619-8

### Geopolitical Conflicts

Groundwater depletion triggers geopolitical conflicts by intensifying competition over transboundary aquifers and shared river basins, turning water scarcity into a security threat.[102] Since transboundary aquifers cross national and regional boundaries, their exhaustion directly threatens global food security and economic stability. [103] Individuals, groups and countries fight over access to and distribution of groundwater using words, laws and weapons. [104] Transboundary water disputes also act as powerful accelerators of the global water crisis. Underground aquifers and river systems cross international borders, allowing upstream nations to build dams or drain shared reserves, harming communities downstream. [105]

## CHALLENGES

### Private Sector Persistence

Private sector utilisation of groundwater is one of the main sources of commercial water withdrawal. Private firms take advantage of weak regulations, or actively lobby against groundwater limits to legally drain aquifers at the expense of the surrounding population's drinking supply. The textile and agricultural industries need water for growing natural fibres and industrial manufacturing processes.[106] Oil, gas, coal, and mining industries both drain and pollute groundwater resources at very high rates and in general, industrial factories use water at an unsustainable level.[107] This causes localised land subsidence and coastal saltwater intrusion. Corporate extraction has pushed many of the world's major aquifers into constant decline.

### Environmental Precarity

Environmental precarity, also known as ecoprecarity, is a state of chronic insecurity and vulnerability caused by environmental damage, climate change, and resource loss. It describes an unstable reality where both human and natural systems face unpredictable ecological threats.[108] Groundwater is the world's largest source of freshwater; hence, it acts as a climate buffer. However, systemic overextraction and contamination are rapidly causing groundwater exhaustion. This exhaustion leads to food insecurity, land subsidence, and seawater intrusion. Groundwater plays an important role

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[102] *(PDF) water conflicts. geopolitical and socio-environmental aspects*. Research Gate. (2026, July). https://www.researchgate.net/publication/408467109_Water_conflicts_Geopolitical_and_socio-environmental_aspects

[103] Robins, N. S. (n.d.). Groundwater scarcity and conflict – managing hotspots. Research Gate. https://www.researchgate.net/publication/270458924_Groundwater_scarcity_and_conflict_-_managing_hotspots

[104] Kuhn, D. (n.d.). The territorial power of appearing and disappearing groundwater conflicts. Geoforum. https://www.sciencedirect.com/science/article/pii/S0016718526003039

[105] *(PDF) water conflicts. geopolitical and socio-environmental aspects*. Research Gate. (2026, July). https://www.researchgate.net/publication/408467109_Water_conflicts_Geopolitical_and_socio-environmental_aspects

[106] *Chapter 2: The global assessment of private sector impacts on water - global institute for water security*. water. (n.d.-b). https://water.usask.ca/valuing-water-global-assessment/report/chapter-2.php#IndustrialSectorsValueChainAnalysisofPracticesExternalitiesandWaterImpactsGlobally

[107] Parent, M. (2023, October 4). *Depleted aquifers: Causes, effects, solutions*. Earth.Org. https://earth.org/depleted-aquifers-causes-effects-and-solutions/

[108] *International Journal of Humanities Social Science and Management (IJHSSM)*. Ecoprecarity and Literature: Reimagining Narrative and Ecology. (2025, November 23). https://ijhssm.org/issue_dcp/Ecoprecarity%20and%20Literature%20Reimagining%20Narrative%20and%20Ecology.pdf

in supporting ecosystems and is essential for their health and resilience.[109]

### Economic and Data Roadblocks

Untapped or overdrawn, groundwater provides 49 per cent of the volume of water withdrawn for domestic use by the global population and around 43 per cent of all water withdrawn for irrigation, watering 38 per cent of the world's irrigated land.

Groundwater abstraction has played a major role in accelerating food production and food security globally since the 1960s. Of more than 500 of the world's largest cities, more than half have groundwater as part of their water portfolio. More than 80 per cent of large cities in the Middle East, South Asia, and Central Asia rely on groundwater as their main source. [110]

Thus, groundwater is a key economic asset in a country's portfolio to reduce poverty and promote resilient and equitable growth. Groundwater being a key economic factor in the development of several nations, groundwater extraction cannot come to a sudden halt.

The UN Summary Progress Update 2021 on SDG 6 raises the issue of the lack of groundwater data and groundwater monitoring initiatives, emphasising that groundwater monitoring is a 'neglected area'. Data roadblocks interfere with monitoring the current state of aquifers, which may lead to inaccurate measurements of the current level of depletion, recharge or pollution.

## CASE STUDIES

### Plachimada, Kerala

In March 2000, Coca-Cola, under its Indian subsidiary Hindustan Coca-Cola Beverages Private Limited (HCCBPL), commenced operations at its bottling plant at Plachimada, in the southern state of Kerala.

The Kerala State Pollution Control Board (KSPCB) granted the company a permit to produce 561,000 litres of beverage per day, but about 2 million litres of water per day were extracted.

Within six months of commencement, the villagers complained that the water was unsuitable for drinking or cooking; it had turned milky white and was brackish. Several villagers complained of unusual stomachaches, while farmers complained of wells emptying unusually fast and crop yields decreasing. Corpwatch India, a public interest group, found that there were high levels of calcium and magnesium in the water, caused by excessive extraction of water.

On April 22, 2002, the Anti-Coca-Cola People's Struggle Committee began its protest against the plant, with over 1500 people, mostly Adivasis, demanding the immediate shutdown of the plant.

After multiple protests by the samithy and a judicial battle, the Perumatty Panchayat decided not to renew the license of HCCBPL. The secretary of the Panchayat cancelled the license, stating the following reasons: excessive exploitation of the groundwater by the company, environmental problems due to the presence of hazardous and

[109] *(PDF) groundwater and ecosystems: Understanding the critical interplay for sustainability and conservation*. Research Gate. (2024, March). https://www.researchgate.net/publication/379044910_GROUNDWATER_AND_ECOSYSTEMS_UNDERSTANDING_THE_CRITICAL_INTERPLAY_FOR_SUSTAINABILITY_AND_CONSERVATION

[110] Worldbank. (n.d.-q). https://documents1.worldbank.org/curated/en/099257006142358468/pdf/IDU0fb2550de013100434708d920a3e3bec6afb1.pdf?intcid=ecr_hp_headerQ_en_ext

toxic substances in wastes emitted by the company, and a scarcity of drinking water. [111]




The protest in front of the plant at Plachimada, Kerala
Source: Flickr

### Metropolitan Phoenix

Arizona State University researchers reported rapid and accelerating losses of groundwater in the basin's underground aquifers between 2002 and 2024. Some 40 million people rely on water from the aquifers.[112] The Colorado River is drying up, and groundwater aquifers are running dry at a much faster rate than its surface water.

The Valley is adding developments at a rapid pace, and has had to stop some developments that began building without securing a 100-year supply of water, as Arizona law requires. Additionally, the boom in data centres coming to the city will drain water resources.

Farming accounts for more than 80% of total water use in southeastern Arizona, the study noted, and the 15 most water-intensive crops account for 71% of all crops.[113]

Many farms use vast amounts of groundwater in part because they use a water-intensive type of irrigation known as flood irrigation (or sometimes furrow irrigation), a technique where water is released into trenches that run through crop fields. The long-standing practice is typically the cheapest option, but it can lead to more water loss and evaporation than other irrigation techniques, such as overhead sprinklers or dripping water from plastic tubing.

### The Nubian Sandstone Aquifer System (NSAS)

The Nubian Sandstone Aquifer System (NSAS) is the world's largest known fossil water aquifer system. It is located underground in the eastern end of the Sahara Desert and spans the political boundaries of four countries, Libya, Egypt, Chad and Sudan, in northeastern Africa.

The NSAS´s reserves are equivalent to about 500 years of Nile River discharge. While the quantity of usable groundwater is plentiful, it is also finite because the fossil reserves are not an active part of the surrounding hydrological cycle, meaning that the water cannot be replenished. It can only be used once.

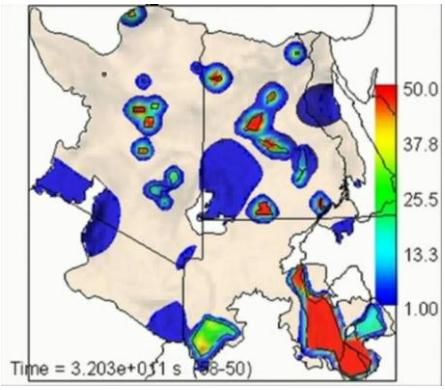
To meet the increasing water demands in Egypt, it has resorted to groundwater as the sole source for irrigation of barren areas. The Gravity Recovery and Climate Experiment (GRACE) mission studies the aquifers' storage changes. The average depletion rate over the Sinai Peninsula was estimated at 0.64 cm/year, and the depletion rate over the Nile Delta aquifer was 0.32 cm/year.

During the investigated period (2003–2021), the extracted groundwater quantity from the Nubian aquifer in the Western Desert is estimated at nearly 7.25 km^3. This reflects the aquifer

[111] Ritimo. (2025, October 23). *The plachimada struggle against Coca Cola in Southern India*. ritimo. https://www.ritimo.org/The-Plachimada-Struggle-against-Coca-Cola-in-Southern-India
[112] NASA. (2026, August 12). *Arizona's declining groundwater - NASA science*. NASA. https://science.nasa.gov/earth/earth-observatory/arizonas-declining-groundwater-154567/
[113] L'Heureux, T. (2025, September 27). *Arizona is losing groundwater at an alarming rate, study shows*. Phoenix New Times. https://www.phoenixnewtimes.com/news/arizona-is-losing-groundwater-an-alarming-rate-study-shows-21915906/CommissionCommission

exposure for extensive water pumping to irrigate newly cultivated lands.[114]

It is possible that sometime in the future the NSAS will run dry, making it increasingly important to manage the aquifer by understanding the dynamics of the groundwater system.[115]




Modelling results showing the effects of pumping over the course of 200 years with colour scaling in meter decrease.
Source: IAEA

## CONCLUSION

The impacts of commercially induced depletion of groundwater by private enterprises, the agricultural sector and other industries are severe economic and social issues, such as an immediate food production collapse, a severe drinking water crisis, land subsidence and geopolitical conflicts.

Legal frameworks and laws on transboundary aquifers, such as the Franco-Swiss Genevese aquifer, maintain the depletion level of aquifers and prevent conflicts. The UN draft articles on Transboundary Aquifers adopted by the International Law Commission and introduced by the UN General Assembly serve as the foremost law that guides countries to manage their shared underground water resources.

However, as these principles are non-binding, corporations in some regions take advantage of weak legal frameworks and diminish regional aquifers much more than permitted for their commercial use.

Other challenges, such as environmental precarity, make it difficult for countries prone to droughts and harsh conditions to preserve groundwater, which acts as the primary source of water.

There is also a severe lack of data when it comes to the amount of water extracted for commercial use, as many countries lack the infrastructure to implement water monitoring, further risking the needs of the community, especially in times of scarcity.

[114] Shalby, A., Emara, S. R., Metwally, M. I., Armanuos, A. M., El-Agha, D. E., Negm, A. M., & Gado, T. A. (2023, April 20). *Satellite-based estimates of groundwater storage depletion over Egypt*. Environmental monitoring and assessment. https://pmc.ncbi.nlm.nih.gov/articles/PMC10119068/

[115] The water is ancient, the secrets are many. (n.d.-g). https://www.iaea.org/newscenter/news/water-ancient-secrets-are-many